

Performance Evaluation of Sonic Fire Fighting System

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Abstract—Water and chemical based methods are the traditional fire extinguishing techniques. There is an urgent need to develop an extinguisher that can overcome the problem of residues and wastage of water. The best solution and alternative to these traditional methods is sound waves. It is the cleanest method in extinguishing flames. The sound waves will displace the original source to a different position, leading to reduction in oxygen and thus extinguishing the fire. The major focus of the project will be to analyze and evaluate the effect of various frequencies of acoustic waves on flames. It will also involve the study of behavior of sound wave propagation in the truncated designed collimator and various environmental conditions. The low frequency sound wave device suggests a model which includes an acoustic wave generator, power amplifier, subwoofer, and collimator. The testing has been done on the sources of class A and Class B fires. The frequency of 63.1 Hz has been identified for extinguishing the fire. The Collimator design has been modified in order to achieve better convergence and faster time of operation. The modification of Collimator has made it possible to extinguish small as well as medium fires. The variation of high and low pressure in sound waves when combined with a high flow air velocity causes disruption at the flame boundary in the air-fuel ratio, and thus leading to flame extinction.

Keywords—extinguisher, alternative, acoustic, flame extinction, collimator

I. INTRODUCTION

The principal motive behind above mentioned exploration work is to supersede the present use of chemical and water based extinguishing methods by sound waves, thus leading to optimize usage of precious assets along with the protection of health of individuals and apparatus damage resulted by chemicals. Fire is the outcome of quick oxidation of a material in an exothermic reaction of combustion, leading to liberation of light, heat, and various other products of reaction. The presence of oxygen is responsible for survival of fire. When the temperature of the combustible material or fuel along with presence of oxygen comes in contact with an amount of heat, just above its threshold, a chain reaction takes place which ultimately

leads to its ignition. The four significant components of fire are: Fuel/Combustible component, Heat, Oxygen, and chemical reaction. This combination of fire is known as the fire tetrahedron.

Acoustic waves can be divided in three different classes:

1. Audible waves:

These waves are heard by human ears. Their range of frequency is extending from 20 Hz to 20 KHz. It may vary from individual to individual as per her/his lifespan. Children and young generation can hear sounds of up to 20 KHz frequencies whereas elderly person audible range is reduced and is less than 20 KHz.

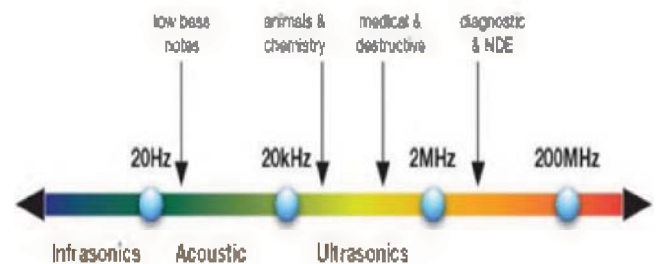


Fig. 1. Frequency spectrum of sound waves

2. Infrasonic waves:

Infrasonic wave involve frequencies less than 20 Hz. Human ears cannot hear such types of waves. The vibration of huge bodies like the earth results in production of such waves.

3. Ultrasonic waves:

Ultrasonic wave have frequencies more than 20 KHz. These waves can be heard by bats but not by human ears. The bats hear influxes of frequencies up to 50 to 60 KHz and even deliver such waves.

Acoustic waves to quench fire. This idea offers the cleanest approach to stifle fire [16].

The commitment of this exploration based work is related with the presentation and execution of environmental-friendly smart fighting fire extinguisher utilizing waves of sound as origin. The accomplishment of this goal, comprehension and contemplating the involvement of natural science behind the acoustic-flame communication for applications in practical sense, recognizing the frequency range in which an experiment can be carried out, deciding the best possible area, distance and where this framework would be achievable and involve the covering of all the fundamental topics.

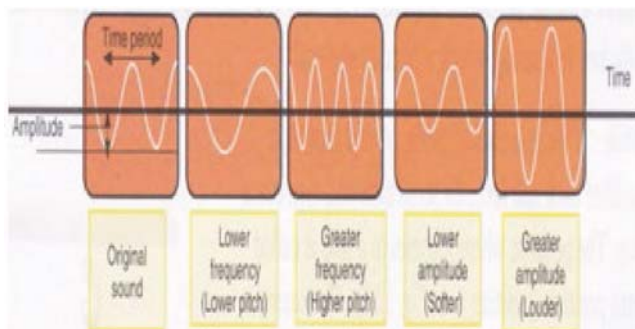


Fig. 2. Pictorial representation of sound waves

This paper is divided into eight sections. Section I deals with the aim of the project as well as the research focus. Section II discusses the basic principle involved. Section III highlights the approach followed for the project. Section IV involves the experimental prototype utilized for efficiently extinguishing the fire. Section V includes the limitations encountered while building the prototype. Section VI details about the Result and discussion which leads to drawing of conclusion in Section VII. The applications of the eco-friendly model are discussed in last Section V

II. BASIC PRINCIPLE INVOLVED

These acoustic waves produce high pressure and low-pressure areas in the medium like oxygen, thus causing the displacement of oxygen from its initial place to a new one. This action of the sound waves is the basic principle of working of sound fire extinguisher.

As per Gay Lussac's law, "the pressure of a given amount of gas held at a constant volume is directly proportional to the Kelvin temperature". As per the law the mathematical relation is expressed as depicted in equation given below:

$$P \propto T \quad (1)$$

As per the relation, expressed by the mathematical formula diminishes, the conditions of temperature, likewise reduces. As indicated by law, discussed above, the waves of sound will aid in diminishing the temperature conditions of source and thus consequently smothering the origin of the fire.

TABLE I. HEARING RANGE OF DIFFERENT ANIMALS IN HERTZ

Animals	Hearing range(Hz)
Humans	20-20000
Bats	2000-110,000
Elephant	16-12000
Dog	67-45000
Cat	45-640000
Rat	200-76000
Opossum	500-64000
Chicken	125-2000
Horse	55-33,500

The displacement property of the waves of sound, utilized above indicates that the oxygen medium is dislodged from the area of the combustion response (fire) and subsequently leading the response to die down.

With reference to figure1 we observe that our model, working at 63.1 Hz is within the hearing range of most of the animals and hence is not harmful. The frequency of 63.1 Hz comes in the acoustic range of the frequency spectrum.

The hardware model is designed in such a way that it centralizes the waves of sound in one direction specific area as opposed to diverging of the waves.

III. APPROACH

The proposed approach will be the following:

- Introducing a smart way to extinguish fire. In our case, we will be using sound waves as a means to do so.
- To identify the range of frequency within which the experiment could be performed.
- To understand and study the physics related behind the sound-flame communication.
- To try and determine the range and area of fire that can be smothered utilizing this method [4].
- To verify whether this technique is feasible in the real world applications or not.

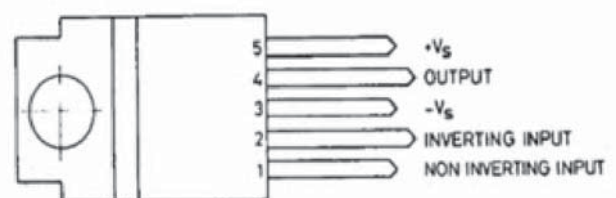


Fig. 4. TDA 2030 IC

For the verification of its impact on the fire as stated by the previous practical experiments. The work of this range of sound waves will be performed on different classes of fire based on their corresponding types of sources like wood based, electrical based and metal based etc. Additional point to be verified in this experiment is to observe for any change in the frequency of the acoustic wave, if the intensity of the

sound remains same, for maximum extinguishing of different classes of fire.

For this experiment to work appropriately and deliver precise results, a collimator is to be intended for the assembly of all the waves of sound originating from the origin, subwoofer in this situation, to center at a specific direction, where the source of fire will be placed.

IV. EXPERIMENTAL PROTOTYPE

This model includes 3 sections for quenching fire; first section involves the employment of tone generator for generating the signal. The Second section involves the amplification procedure that is being utilized through the assistance of TDA 2030 ICs. Ultimately, the subwoofer is utilized for generation of acoustic signal in the third section.

The TDA 2030 is a class AB amplifier IC having a maximum output power of 20 Watts; this amplifier board has three of TDA2030 ICs in a cascaded connection for giving sufficient output power to drive the subwoofer. Output of the amplifier circuit goes to the transducer (subwoofer in our case) for converting the electrical power into acoustic power required for the project.

The TDA 2030 has some handy features like short circuit protection and thermal shutdown which makes it useful in a lot of situations.

For this model to work appropriately and deliver precise outcomes, collimator is intended for assembling all waves of sound originating from the origin (a subwoofer in this situation), to center towards the direction of the fire source.

V. LIMITATIONS

The plan of stifling flames through sound faces various difficulties regardless of the exploration going on. A portion of above mentioned plans are referenced underneath:

A. Noise

This model gives rise to a immense measure of commotion that could harm the physical strength of the personnel working with equipment. Continuous subjection to commotion can likewise prompt hearing problems. Different impacts include high blood pressure, stress, sleep related problems etc.

B. Range

The major difficulty that needs attention is the decrease in dropping of waves of sound when it is communicated to a significant distance, in order to quench fire over a wide area.

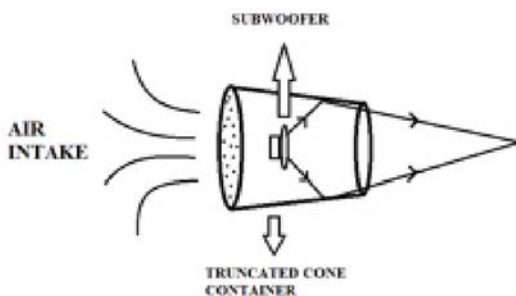


Fig. 5. Collimator design and flow of sound

For making the waves of sound progressively compelling, the model being proposed utilizes the structure

of a vortex cannon, which frames a donut shaped gaseous projectile when a stream of air goes by it.

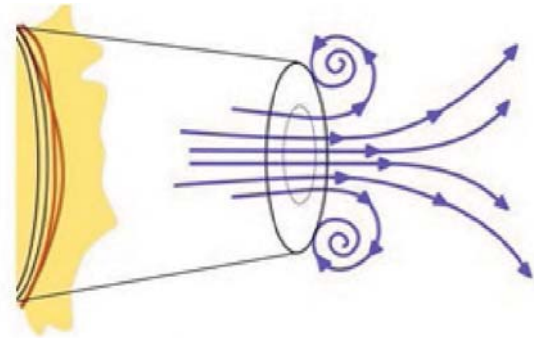


Fig. 6. Vortex Cannon

C. Miniaturization of design equipment

The present structure of the model proposed is very massive and requires alterations so as to make it light in weight. The dimensions of a model are huge and hence cannot be installed inside buildings. Personnel need to carry out the model alongside them at all times.

D. Output

The outcome of the model proposed is not palatable. This model smothers a fire at a small separation that isn't worthy of compliments. The adequacy of a certain frequency needs to be in extent with the width of the fire displaced.

VI. RESULT & DISCUSSION

The output from the subwoofer as observed through an oscilloscope has been shown in the figure 7. The average output frequency as observed comes out to be around 63 Hz. The waves produced at this frequency are optimal for extinguishing a short range fire.

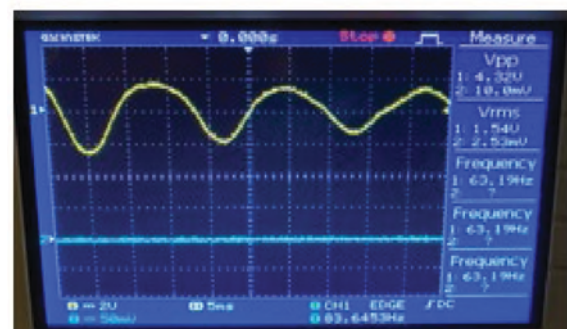


Fig. 7. Waveform output

Further, experimentation was done taking different materials of different sizes into account. The experimentation is classified into two classes. One includes small range fires and the other includes medium range fires. The performance of the experimentation done was evaluated and graphed as in figures 8 and 9.

Here, kerosene oil was taken in both the experimentations. In the small fire range, the quantity of the oil taken was around (8-10ml), while that in medium fire range was around (25-30ml).

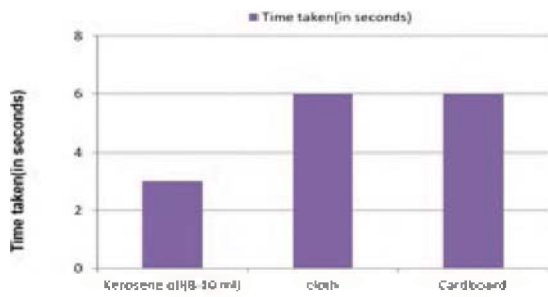


Fig. 8. Time graph for extinguishing different materials (Small fire range)

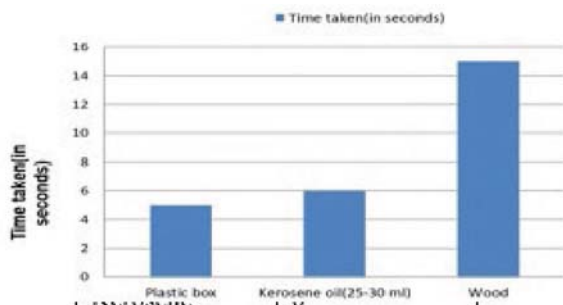


Fig. 9. Time graph for extinguishing different materials (Medium fire range)

VII. CONCLUSION

From practical experimentation, it is observed that only low frequencies of sound waves can be utilized for effectively extinguishing the fire. As per theoretical principle, the large time period (inversely proportional to the frequency) due to the high-pressure and the low-pressure regions of sound wave leads to increase in the time of low amount of oxygen present in the surrounding of fire source. This makes it more likely for the combustion reaction to stop. Hence, the best range of frequencies came out to be in the range of **50 Hz to 70 Hz**.

The maximum distance between the acoustic fire extinguisher and the social fire is around 0.75 m. Several reasons factor the amount of this distance:

- The maximum Output power utilized by the subwoofer.
- Structure of the collimator
- Intensity of the fire
- The input signal given

TABLE II. TIME TAKEN TO EXTINGUISH FIRE ON DIFFERENT MATERIALS

Object Tested	Time taken (in seconds)
Plastic Box	5
Kerosene Oil(25-30 ml)	6
Wood	15
Kerosene Oil(8-10ml)	3
Cardboard	6
Cloth	6

The maximum output power of the project was made for the purpose of researching the feasibility of extinguishing fire to some waves and thus, high-power components weren't utilized for increasing the output power. The

collimator is a customized container housing the subwoofer of the project and as such can be made in several shapes Made from various materials. From practical experimentation it was found out that truncated cylinder shape converges the sound waves with high efficiency. Therefore, truncated cylinder shape is utilized for the collimator of sound fire extinguisher. The input signal given through the Tone Generator Is oscillating with the frequency of 63 Hz, which is a relatively efficient frequency range utilized for extinguishing fire.

VIII. FUTURE SCOPE

The concept of Sonic Fire Extinguisher can be really helpful in applications such as extinguishing fire in outer space. Though fires do not usually occur in outer space due to the lack of oxygen, it can occur in spacecraft due to any malfunction. Fires occur differently in space than they do on Earth and hence astronauts deal with it in a different manner. The best way to fight a fire in space is to avoid it from generating.

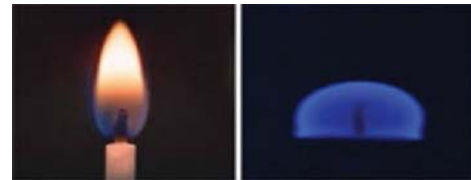


Fig. 10. Flame of fire acting differently on earth and in space

If a fire occurs on a spacecraft, the astronauts would follow a 3 step procedure. Firstly, they would turn off the ventilation system, second, they would cut off the electrical supply to that unit. Lastly, they would use the fire extinguishers to put out the fire. The chemicals used in the extinguishers which are used, have a high chance of damaging the equipment and surroundings of the inside the spacecraft, which further needs to be vacuumed. The sonic fire extinguisher can be a potential solution to this problem. Apart from being safe, it will be cheaper and more reliable than the traditional extinguishing elements. Apart from this, the design and purpose of our proposed model can be an epitome in overcoming the disadvantages offered by the traditional

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